

Coupling-gated cortical coherence: what a field theory of conscious unity must assume, and what it predicts

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Abstract

A field account of cortical coordination holds that spatially distributed neural processes are coupled in part through the endogenous extracellular electromagnetic field. We ask what such an account must assume to reach a globally consistent, self-referential state, and what it predicts before it gets there. Its physical core is a noisy mean-field phase model whose coupling drifts through a synchronization threshold. If recovery of coherence after sleep is gated by that crossing, it is not free to take an arbitrary time course. For as long as the phase distribution tracks the stationary branch, it must show a square-root onset and a von Mises concentration–coherence relation with no fitted shape parameter. The tracking condition is part of the prediction, and the simulations measure its cost: ramp speed sets the apparent onset exponent, so an exponent fitted to a driven recovery trace reports the drive as much as the mechanism. A coupling plasticity that descends a squared-drift objective removes most of the objective available to it and keeps the system coherent, while driving within-cluster coupling far below between-cluster coupling—away from the

structure it is asked to learn, and further from it than from almost every relabelling of the nodes; a matched random update of the same step size does neither. Descending that functional does not produce representational learning, and its optimum is opposed to the target by a conservation law rather than by accident. A Lean 4 formalization composes finite capacity, Landauer’s bound, the noisy Kuramoto transition, sheaf gluing and Banach’s theorem into one conditional theorem, isolating the eight hypotheses it consumes and the two structures it takes alongside them, whose class fields carry physical content of their own; one witness assembled from distinct toy components satisfies all eight at once. We defend as a philosophical commitment, not a theorem, the identification of the glued section with the unity of a conscious episode and of its reflexive fixed point with a minimal self. Testing the account requires independent calibration of effective cortical coupling and an awakening experiment separating slow coupling change from alternative recovery mechanisms.

1 Introduction

Neural activity changes across sleep, awakening and anesthesia, but a change in synchrony does not by itself identify the physical interaction responsible for it. A field account makes a stronger proposal: spatially distributed neural processes are coordinated partly through the endogenous extracellular electromagnetic field, and changes in the medium can change that coordination (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Whether that mechanism is sufficient to support the collective state such accounts require remains an empirical question.

Such an account has three requirements, and they order what follows. Coupling must be strong enough to sustain coherent collective activity; the

13 local descriptions that activity carries must be compatible enough to glue
14 into a single one; and that glued state must support an accurate model of
15 itself. The first requirement is physical and testable with current methods,
16 and most of this article is about testing it. The second and third are where
17 the mathematics is, and where the assumptions accumulate.

18 The first requirement admits a definite claim. If cortical coupling is slowly
19 gated—if recovery of coherence after sleep is driven by a coupling parameter
20 drifting through a synchronization threshold—then the recovery is not free
21 to take an arbitrary time course. For a recovery slow enough that the phase
22 distribution tracks the stationary branch, the theory fixes a square-root onset
23 and a one-parameter relation between phase concentration and coherence,
24 with no fitted shape parameter. That is a constraint an experiment can
25 meet or violate. How slow is slow enough is itself part of the claim, and
26 our simulations bound it from the other side, by showing which parts of the
27 constraint survive finite populations and finite ramp speeds and which do
28 not.

29 The contribution has four parts. The first is the coupling-gated recovery
30 hypothesis itself: a stationary coherent branch with square-root onset, a von
31 Mises concentration–coherence relation, and an experimental protocol with
32 failure conditions declared in advance (Sections 2 and 3). The second is a
33 constraint on how such data may be analyzed. The apparent onset expo-
34 nent measured from a finite ramp is a function of ramp speed across the
35 sweep of Section 4, so an exponent fitted to a recovery trace measures the

36 ramp at least as much as the mechanism—and that holds for any analysis
 37 reading a critical exponent off a driven transition, not only for ours. The
 38 third is a negative result about plasticity: a coupling update descending
 39 a squared-drift objective removes most of the objective available to it and
 40 preserves coherence, while moving *away* from the environmental structure
 41 it is expected to learn—below almost every node relabelling, and beyond
 42 what a matched random update of the same step size does. Descending a
 43 squared-drift functional does not produce representational learning, and the
 44 direction of that failure follows from a conservation law rather than from the
 45 particular run. The fourth is a machine-checked composition (Section 5) in
 46 which finite information capacity, Landauer’s bound, a predictive dissipation
 47 bound, the noisy Kuramoto transition, sheaf gluing and Banach’s fixed-point
 48 theorem yield one conditional theorem, with the eight hypotheses that com-
 49 position consumes isolated and named; one witness assembled from distinct
 50 toy components satisfies all eight at once, so their conjunction is not empty.

51 A prose table of assumptions would list the same eight items, so the fourth
 52 contribution needs a word about method. What the Lean development adds
 53 is that it makes the list non-negotiable. The characteristic failure in an
 54 argument of this shape is not a false step but a silent change of subject:
 55 a theorem about a stationary density read as a claim about a trajectory,
 56 an inequality about a register read as a claim about learning, agreement of
 57 phases read as compatibility of representations. Type-checking forbids each
 58 of those substitutions, and an assumption needed to make a step go through

59 appears as a hypothesis in the term rather than as a word in a sentence. The
60 last of them is not merely forbidden but refuted by a constructed object: the
61 development carries a cover at a single common phase whose local densities
62 disagree where its patches meet, satisfying every other requirement of the
63 gluing theorem and failing compatibility alone (Section 5.2). That object
64 is what a prose table of assumptions cannot carry. Formalization does not
65 make the assumptions true; it makes them impossible to lose.

66 The physical model and the formal composition belong in one article be-
67 cause they are two halves of one commitment. The coupling-gated prediction
68 is worth testing under this framework, rather than as one more synchroniza-
69 tion model, only because the coherent state it produces is claimed to do
70 further work: to support a globally consistent representation and a reflexive
71 fixed point that we identify with phenomenal unity and a minimal self.

72 The distinction that does that work is between phases agreeing and de-
73 scriptions fitting together, and it is concrete. Take two cortical regions whose
74 receptive territories overlap, each carrying a probability distribution over
75 what occupies the shared part. Their oscillators can be locked to the same
76 phase at every instant while one region's distribution on the overlap places an
77 object left of fixation and the other places it right. The phases agree; the de-
78 scriptions do not, and no single distribution over the shared territory restricts
79 to both. A theory that identifies unity with synchrony cannot mark that dif-
80 ference, and so cannot separate a brain whose regions oscillate together from
81 one whose regions are about the same world. The sheaf condition marks

82 exactly it: local states glue to one global section when their restrictions to
 83 shared regions coincide, and otherwise not at all. Coherence enters as the
 84 physical condition under which compatibility can be maintained, not as a
 85 substitute for it, and the self-prediction map then asks the further ques-
 86 tion of whether the glued state contains an accurate model of itself. That
 87 identification is a commitment, not a theorem, and Section 6 states it, ar-
 88 gues for it, and says what would count against it. Its separation from the
 89 physical tests is what makes both assessable: the cortical coupling hypothe-
 90 sis and the recovery prediction can fail while the mathematical composition
 91 stands, and the composition can stand while the identification is wrong. The
 92 established ingredients (Landauer, 1961; Still et al., 2012; Sakaguchi, 1988;
 93 Banach, 1922) therefore appear here for their biological composition, not for
 94 any new fundamental physical law.

95 **2 A coupling-gated model of cortical coher-** 96 **ence**

97 **2.1 Dynamics and stationary solutions**

98 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units, with angles treated as dimensionless. The uniform mean-field convention is $K_{ij} = K/N$. Frequency heterogeneity and phase diffusion are separate parameters; a frequency-spread width cannot be substituted for D when applying the identical-frequency stationary theory.

For identical frequencies in their rotating frame, the assumed stationary density is

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}. \quad (2)$$

The I_n are modified Bessel functions, and ψ is the collective phase. Lean verifies normalization, the density's circular order parameter, and the self-consistency equation. At $K \leq K_c = 2D$, the only nonnegative self-consistent order parameter is zero. At $K > K_c$, exactly one positive solution coexists with zero; the positive branch is continuous at threshold and strictly increasing with K .

The near-threshold expansion gives

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

This is a result about the stationary solution branch. Of the coherent branch we prove existence, uniqueness, continuous emergence at threshold and strict growth in K , but not that it is dynamically selected. The dynamical mean-field limit—propagation of chaos—is the missing link between the finite sys-

tem and the stationary description (Sznitman, 1991), and no theorem in this development closes it. Exchange symmetry alone does not justify factorization of the joint law, particularly for an ensemble with an unpinned collective phase.

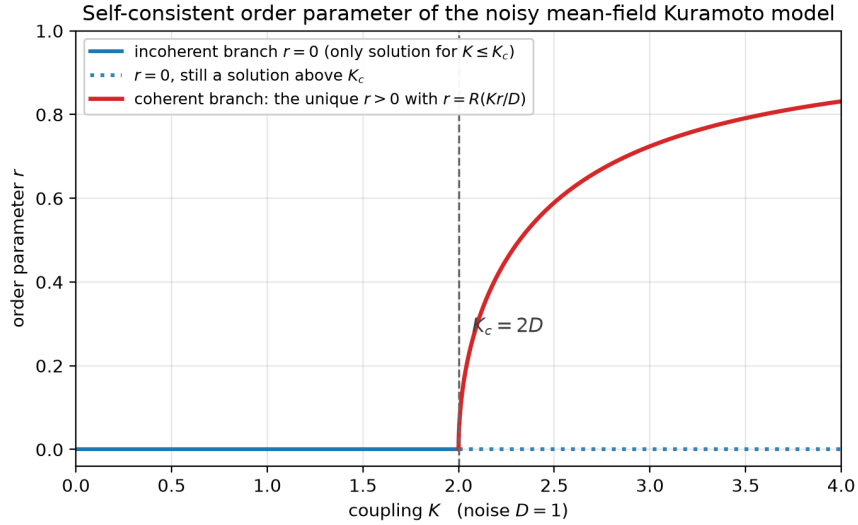


Figure 1: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the same integral expressions used in the formalization; it illustrates the static theorems and does not establish dynamical selection.

A spatial version replaces the finite sum by an integral against a kernel $K(x, y)$. We normalize the substrate measure to total mass one, so the mean-field strength is a kernel average, not a quantity that grows with the arbitrary measure of the substrate. Reduction of a heterogeneous spatial kernel to a scalar K is an additional modelling step, and the threshold in Eq. (2) does

126 not automatically apply to every spatial or signed network.

127 **2.2 Physical identification and calibration**

128 Regular mesh refinement gives convergence of a discrete coupling-energy
129 quadrature to its continuum integral: scalar energy convergence, not conver-
130 gence of trajectories and not a construction of a physical kernel. The model
131 posits a continuum kernel and separately identifies its cortical realization
132 with endogenous electromagnetic coupling. The formal predicate requires
133 a jointly continuous kernel, a normalized substrate, a mean-field strength
134 equal to its spatial average, and no single site providing more than half of
135 that mean-field coupling. Toy witnesses satisfy these conditions; none is a
136 model calibrated to cortex.

137 Extracellular fields can modulate spike timing (Fröhlich & McCormick,
138 2010; Anastassiou & Koch, 2015), and field-based models have been proposed
139 for integrating distributed neural information (Pinotsis & Miller, 2022; Mc-
140 Fadden, 2020). These observations motivate an effective interaction without
141 establishing its magnitude relative to phase diffusion, its spatial averaging
142 rule, or its causal role in experience; changes in phase organization under
143 propofol (Bastos et al., 2021) likewise do not measure a crossing of the model’s
144 coupling threshold.

145 The physical-unit conversion is the sharpest open limitation of the frame-
146 work. An estimate proportional to $N\Delta tf$, using population size, a spike-time
147 shift and oscillation frequency, is dimensionless. It cannot be compared di-

148 rectly with an inverse-time diffusion parameter. A conversion such as

$$K_{\text{eff}} = \gamma N s E f \quad (4)$$

149 requires a rate calibration γ , where s is the time shift per unit field and E
150 the field amplitude. Supplemental Material sets out why γ is not recoverable
151 from the other factors: it states how a per-cycle phase advance accumulates
152 into a drift rate, and the choice that would let it cancel removes the very
153 ratio the threshold test is about. The saved field-strength calculations are
154 conditional on this calibration and on the chosen spatial aggregation; they do
155 not independently establish $K_{\text{eff}} > 2D$ in cortex. Determining the interaction
156 kernel and phase diffusion from the same physical system is necessary before
157 the threshold becomes a quantitative biological claim; Section 3.2 makes that
158 measurement the first item of the proposed experiment.

159 **3 Recovery after awakening: a conditional** 160 **prediction**

161 Sleep-associated changes in extracellular space and clearance motivate a pos-
162 sible slow control of coupling (Xie et al., 2013). We hypothesize that recovery
163 after awakening includes a change in effective coupling $K(t)$, mediated by
164 extracellular geometry, on a slower timescale than phase dynamics. Neither
165 the monotonic relation between geometry and coupling nor its role in hu-

man sleep inertia follows from the cited physiological observation; the model requires both to be measured.

If $K(t)$ rises through K_c and the phase distribution tracks the coherent stationary branch, Eq. (3) predicts

$$r(t) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad t > t_0, \quad \dot{K}(t_0) > 0. \quad (5)$$

The ideal stationary description has a subthreshold interval followed by continuous onset. Saturation requires a specified subsequent coupling trajectory; monotonic coupling alone does not determine a logistic or any unique sigmoid. The onset time is fixed by $K(t_0) = K_c$ only if $K(t)$ is independently known. When coupling is inferred from the same coherence trace, this is not a parameter-free prediction of latency.

Critical slowing down limits the adiabatic prediction of Eq. (5) near threshold, and a finite population carries nonzero fluctuating order below it. An observable onset therefore requires a predeclared threshold above the finite-size floor, and a finite-rate trajectory need not show an infinite-slope cusp. Section 4.1 quantifies the limitation.

3.1 Distributional and temporal tests

The stationary model makes a second prediction: phase concentration and coherence obey $r = I_1(a)/I_0(a)$, with no fitted shape parameter. Its applicability depends on the stationary-density assumption and the observation

185 model, and the relation is common to the whole von Mises family, so agree-
 186 ment with it identifies neither ephaptic coupling nor this framework against
 187 alternative synchronization models. Within the same ansatz $a/r = K/D$ for
 188 $r > 0$, so the limiting threshold ratio follows from the density relation instead
 189 of testing it, and ratios near the finite-size floor are especially unstable.

190 The estimators must be independent, which is the constraint most easily
 191 violated in practice. Estimating a by inverting the Bessel ratio from r would
 192 enforce the relation being tested, and linear statistics on wrapped phases
 193 estimate neither quantity.

194 **3.2 A protocol for the awakening experiment**

195 The requirements above define a design rather than a single measurement,
 196 and the design is where the content of the hypothesis lives. An adequate test
 197 needs all six of the following.

- 198 **1. Coupling and diffusion in the same units.** Estimate K and D
 199 from the same cortical regime, in inverse-time units, separating phase
 200 diffusion from quenched frequency spread and from observation noise.
 201 Without this the threshold $K > 2D$ is not an evaluable statement
 202 about cortex.
- 203 **2. Geometry calibrated to coupling.** A diffusion-MRI observable or
 204 an astrocytic signal is an indicator of extracellular geometry, not a mea-
 205 surement of K . Its mapping to effective coupling must be established

206 separately, with its own uncertainty.

207 **3. Phase estimation at adequate spatial resolution.** At least 100
208 spatial samples per estimate. Estimate r by the circular resultant
209 length and a by an unweighted regression of log-density on the co-
210 sine of phase relative to the circular mean. With fewer sites, pooling
211 over time changes the estimand to a spatiotemporal distribution and
212 must be stated as such.

213 **4. An independent recovery endpoint.** Behavioral recovery measured
214 separately from the phase trace, so that latency is not defined by the
215 signal whose latency is at issue.

216 **5. A predeclared onset criterion.** A coherence threshold fixed above
217 the finite-size floor before unblinding, chosen in the knowledge that a
218 finite ramp smears the cusp.

219 **6. Model comparison under matched observation.** Coupling-gated
220 recovery compared against exponential, flexible sigmoidal and other
221 slow-recovery models under identical temporal filtering and observation
222 noise. Latency alone cannot distinguish the mechanism, because many
223 processes create delays.

224 The design fails informatively in three ways: failure of the stationary
225 phase-density relation over a sufficiently broad concentration range; absence
226 of the proposed coupling-geometry association; or a recovery trace better

227 explained by an alternative after finite-rate and observation effects are ac-
 228 counted for. Two apparent failures are not failures. A linear apparent onset
 229 is evidence against the adiabatic approximation only where that approxima-
 230 tion is independently justified, and it does not falsify the field hypothesis
 231 under arbitrary ramp speeds. The absence of a fold in the stationary equa-
 232 tion excludes stationary coexistence of two positive branches in this model,
 233 but does not exclude finite-rate hysteresis or adjudicate theories of ignition.

234 **4 Simulation results**

235 The simulations characterize the finite-size, finite-rate regime that the sta-
 236 tionary theory leaves open, and they bear on whether the prediction of Sec-
 237 tion 3 is measurable at all. Two of them, Sections 4.1 and 4.3, illustrate
 238 methodological issues that also arise in analyses of driven transitions and
 239 dissipation-driven plasticity. Each is a finite run at its sampled parameters
 240 and not a general statement, and none discharges a hypothesis of the com-
 241 position in Section 5. Each run uses explicit seeds and compact saved sum-
 242 maries, and the numerical values below are generated from those artifacts.
 243 The supplement gives parameters, estimators, failure criteria and additional
 244 panels. Identical-frequency runs use phase diffusion rather than a quenched
 245 frequency-spread surrogate, and order plots display the $1/\sqrt{N}$ reference.

246 4.1 Ramp speed sets the apparent onset exponent

247 The dynamic-ramp study (Fig. 2) uses $N = 2000$, $D = 1$ and 32 replicas at
248 four coupling speeds. Escape is the first of three consecutive saved samples at
249 $r \geq 0.2$ after threshold. Only 5/32 replicas escape within the fastest window.
250 The three slower ramps have mean coupling delays 0.1947, 0.0684 and 0.0253,
251 with a fitted delay exponent 0.443. The apparent onset exponent changes
252 from 0.964 at the fastest speed to 0.417 at the slowest.

253 The exponent recovered by fitting a driven transition is a joint function
254 of the mechanism and the drive rate. An onset exponent read off a recovery
255 trace therefore says little about the underlying transition unless the ramp rate
256 is independently known and the finite-size floor is declared, and that holds
257 for any such analysis, not only for one conducted under this framework. The
258 prediction of Section 3 is the first thing it constrains: the stationary square-
259 root law cannot be imposed on an arbitrary recovery trace, and its absence
260 is not by itself evidence against a coupling-gated mechanism. The specified
261 deviation-based critical ramp speed is not bracketed by the sweep.

262 The dynamical-selection control shows escape toward a coherent branch
263 above threshold. At $K = 2.8$, steady mean order is 0.67735 against the
264 static value 0.68270; repeating the ensemble at a halved integration step-size
265 removes most of the residual (halved-step order 0.68021), attributing the
266 bulk of the gap to Euler–Maruyama bias and leaving a finite-size residual
267 of -0.00250 . The fitted growth-rate slope is 0.41505, compared with the
268 theoretical slope $1/2$. The propagation-of-chaos diagnostics show decreasing

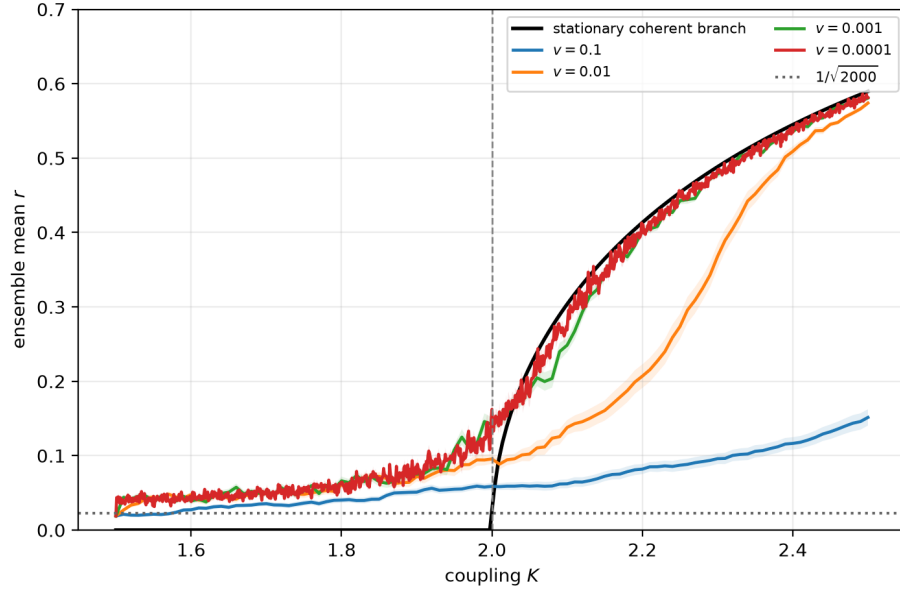


Figure 2: Finite-rate departures from the stationary branch. Ensemble-mean order is plotted against the increasing coupling for four ramp speeds v , with bands of one standard error across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape; the stationary branch is not a universal time course for recovery.

269 subcritical dependence with increasing population size. In the supercritical
 270 unpinned ensemble, however, pair correlation remains substantial because
 271 realizations share a random collective orientation. This negative uncondi-
 272 tional result does not settle a conditional mean-field limit after removal of
 273 that orientation.

274 4.2 Spatial coupling, frustration and calibration

275 A periodic two-dimensional sheet with row-normalized exponential coupling
276 remains coherent in the sampled empirical decay-length band under the de-
277 clared total coupling and frequency spread. At decay lengths 0.1, 0.2 and
278 0.3 mm, steady order is 0.9424, 0.9427 and 0.9427. This is one finite param-
279 eter regime; fixing total coupling while changing range tests fragmentation,
280 not whether cortex supplies that total coupling.

281 A row-balanced Dale-law network shows that row balance alone does not
282 guarantee a floor-level incoherent baseline. In a separately specified weak-
283 synapse regime, a uniform modulatory term produces an operational coher-
284 ence crossing inside the bracket $K_{\text{eff}}/D \in [1.857, 2.110]$ from a geometric
285 grid of ratio 1.136. This bracket contains the model’s own mean-field thresh-
286 old $K_c/D = 2$; the crossing sits below that nominal threshold because signed
287 coupling in this finite-size regime amplifies fluctuations (baseline $\langle r^2 \rangle > 1/N$,
288 making $N\langle r^2 \rangle > 1$). That ratio is dimensionless; carrying it over to cortex
289 needs the rate calibration γ of Section 2.2, which neither this sweep nor the
290 spatial control supplies.

291 4.3 Squared-drift plasticity does not learn structure

292 The predictive dissipation bound of Section 5 bounds nonpredictive memory
293 by dissipation, and it is tempting to read a system that lowers its dissipation
294 as thereby coming to mirror the structure of its input. The bound licenses no

295 such reading: a memory carrying no information at all satisfies it with room
 296 to spare, so lowering dissipation presses toward less representation rather
 297 than toward better-aligned representation. This control exhibits a system in
 298 which the pressure is not merely toward less but away.

299 For fixed phases, consider the squared-drift objective

$$\sigma(K) = \frac{1}{D} \int \left[\omega(x) + \int K(x, y) \sin(\theta_y - \theta_x) d\mu(y) \right]^2 d\mu(x). \quad (6)$$

300 It is a quadratic functional of an affine image of the kernel. The formalization
 301 constructs the bounded drift operator A and its adjoint on the relevant L^2
 302 spaces, obtaining $\nabla\sigma = (2/D)A^*(AK + \omega)$, and the specified gradient flow
 303 decreases this objective at fixed phases. The objective is not generally a
 304 thermodynamic entropy-production rate, which requires probability-current
 305 information. Whether biological couplings follow this flow is an additional
 306 modelling question.

307 The joint control evolves noisy phases with clustered frequencies and
 308 nonzero grand-mean drive, while updating symmetric couplings every 50 fast
 309 steps. Couplings remain nonnegative, hollow and normalized to a fixed to-
 310 tal. The symmetric-edge derivative of $\frac{1}{2} \sum_i v_i^2$ is $(v_i - v_j) \sin(\theta_j - \theta_i)$, so
 311 both endpoints contribute. Resource constraints and evolving phases re-
 312 move the fixed-phase monotonicity guarantee. Each seed runs three arms on
 313 one phase-noise stream: the gradient step, a symmetric random step of the
 314 same Frobenius norm, and a frozen kernel. Denying an implication requires

315 exhibiting its antecedent, so the reported quantity is how much of the ob-
 316 jective the update actually removes: symmetric coupling cancels from the
 317 mean drift, which fixes a floor of 990 beneath the frozen arm’s 1380.6063 to
 318 1382.3448, and the gradient arm removes between 0.6773 and 0.6914 of that
 319 headroom. The random arm of matched step size removes none of it, so the
 320 descent belongs to the gradient. The random arm, however, ends an order
 321 of magnitude less deformed than the gradient arm: successive isotropic steps
 322 cancel, leaving cumulative norm growth of 1.2879 to 1.3056 against the gra-
 323 dient arm’s 12.4094 to 14.6958. Because the random arm barely enters the
 324 sparsification channel, it does not separate the gradient descent from cumu-
 325 lative deformation. The learning rate 0.2 is the rate at which the objective
 326 descends furthest across a sweep from 0.001 to 0.4 evaluated on one of the
 327 three reported seeds, not a rate at which the update is known to converge.

328 Across three seeds, second-half order lies between 0.9167 and 0.9222: co-
 329 herence survives the descent. Alignment with the cluster template is read
 330 from the ratio of mean within-cluster to mean between-cluster coupling,
 331 which is unchanged by rescaling the kernel. That ratio starts between 0.9824
 332 and 0.9948 and ends between 0.3507 and 0.6135, while the random arm leaves
 333 it at 0.9965 to 0.9997; against 2000 node relabellings the final kernel is fur-
 334 ther from the true template than from a fraction 0.9510 to 1.0000 of them
 335 (Fig. 3). The reference for that fraction is the frozen arm on the same seeds,
 336 which reaches 0.6435 to 0.9110 without any update at all, and not a null
 337 median of one half; the increment the gradient step contributes is the differ-

338 ence between those ranges, which at three seeds is small enough that they
 339 nearly meet. Two readouts do not carry the effect and we do not rest it
 340 on them. The initial-minus-final Frobenius distance to the template lies be-
 341 tween -58.0962 and -48.1760 , so that distance grows; but the kernel norm
 342 grows by a factor of 12.4094 to 14.6958 over the same interval, and a fixed
 343 total resource constrains the sum of the entries rather than the norm, so
 344 most of that growth is concentration of mass onto fewer edges rather than
 345 loss of alignment. The edge-level correlation with the template ends between
 346 -0.0362 and -0.0208 , near its starting value: at this sparsity the edge-level
 347 variance swamps a contrast that the block means resolve. The template is a
 348 covariance of shared latent cluster signals, not an input to the update or a
 349 measured covariance of the constant frequencies.

350 The objective descends by most of what is available to it, coherence is
 351 maintained, and within-cluster coupling falls to a fraction of between-cluster
 352 coupling. Coherence is therefore not a proxy for representational alignment
 353 in this regime, and a plateau in the objective establishes neither stationar-
 354 ity of the adaptive process nor a thermodynamic entropy-production rate.
 355 Nor is the direction of the failure accidental. Symmetric coupling conserves
 356 the mean drift, so minimizing $\sum_i v_i^2$ can only equalize drift across nodes,
 357 and equalizing drift rewards coupling between clusters of unlike frequency—
 358 the complement of a template that rewards coupling within clusters of like
 359 frequency. The objective and the template are in this pairing opposed by con-
 360 struction, and that is the point: nothing about a system’s dissipation fixes

361 which structure its slow variables acquire. What fails here is the objective
 362 rather than thermodynamically specified learning at large: an Ising model
 363 whose couplings are annealed degrees of freedom, driven by external fields
 364 that carry the data, memorizes and generalizes under ordinary statistical-
 365 mechanical rules (Corberi et al., 2026). Those couplings equilibrate under
 366 a Hamiltonian containing the data; these descend a squared-drift functional
 367 whose stationary point the data never enters. Which functional the slow
 368 variables follow is what decides whether they acquire environmental struc-
 369 ture.

370 **4.4 The estimation protocol demonstrated on public** 371 **EEG**

372 Item 3 of Section 3.2 specifies estimators whose behavior on real recordings
 373 is not obvious, so we exercise them on public propofol-sedation EEG (Bajwa
 374 et al., 2025). The purpose is to show what the protocol produces and where
 375 it is fragile, not to test the awakening hypothesis, for which these data are
 376 the wrong design.

377 Two fragilities appear immediately. The implemented estimator pools 62
 378 channels over a 100-ms window, so it estimates a spatiotemporal distribu-
 379 tion rather than an instantaneous spatial one—a departure from the protocol
 380 forced by channel count, and one that any scalp implementation will face.
 381 Bipolar and common-reference montages give different residuals, making ref-

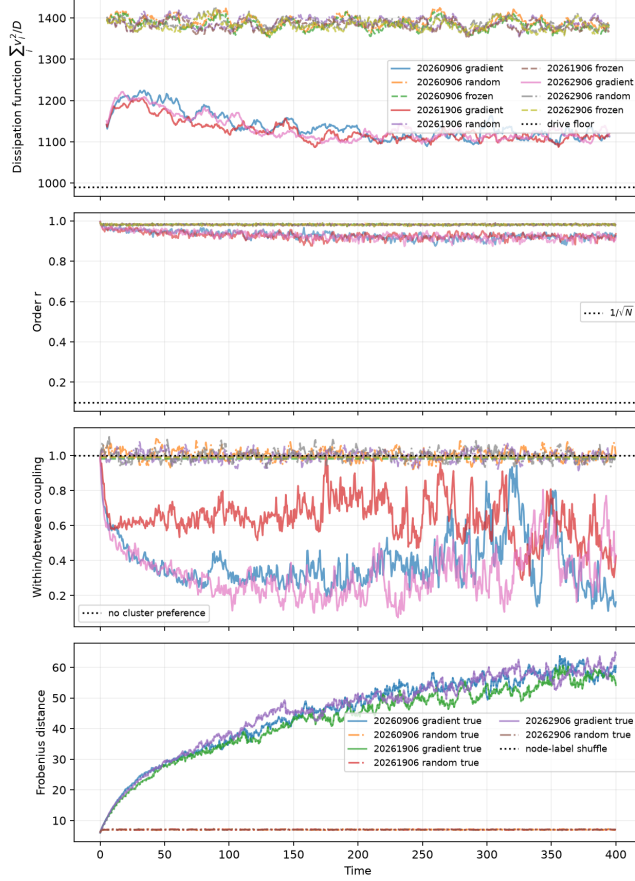


Figure 3: Joint fast/slow dynamics, three arms per seed on one phase-noise stream: gradient (solid), matched-norm random step (dash-dotted), frozen kernel (dashed). First: smoothed squared-drift dissipation against the drive floor the conserved mean drift imposes; only the gradient arm descends. Second: order and the finite-size reference. Third: mean within-cluster over mean between-cluster coupling, a measure unchanged by rescaling the kernel; the gradient arm falls well below the no-preference line while the other arms stay on it. Fourth: Frobenius distance to the true template (solid) and to a node-label shuffle (dotted), which track each other because the kernel norm, not the alignment, is what grows. Coherence persists while alignment is lost. The result limits the proposed connection between dissipation minimization and environmental structure.

382 erencing a material analysis choice rather than a preprocessing detail.

383 The bipolar results are compatible with the Bessel relation over the ob-
384 served concentration range. That range is too narrow to distinguish the
385 curve from its linear approximation at the observed scatter, so the check is
386 uninformative about the relation itself. An informative test needs a broader
387 concentration range and explicit treatment of source mixing, finite samples
388 and temporal pooling; intracranial or high-density recordings across a gen-
389 uine state transition are the natural setting.

390 5 Thermodynamic constraints and the for- 391 mal composition

392 The synchronization model of Section 2 is one link of a longer conditional
393 construction, whose nine nodes are:

394 **n1** a finite phase space has bounded information capacity; **n2** a
395 broken symmetry forces a boundary, where the field leaves its vac-
396 uum manifold; **n3** the boundary erases, and erasure costs heat;
397 **n4** that dissipation bounds nonpredictive memory; **n5** discrete
398 couplings coarse-grain to a continuum limit; **n6** a specified phys-
399 ical field realizes that limiting coupling; **n7** above threshold the
400 field carries a unique coherent order parameter; **n8** compatible
401 local states glue to a unique global section; **n9** that section is the
402 unique fixed point of a self-prediction map.

403 Node-level results are established or cited; what the composition consumes
404 are the eight implications between them. Figure 4 shows where they sit and

405 what kind of assumption each one is, and Table 1 states them, classified by
 406 their role in the composition rather than by their plausibility. In particular,
 407 finite capacity does not imply a symmetry-breaking boundary: n_1 and n_2 are
 408 independent roots, even though the composition can accept an implication
 409 between them as an argument.

410 5.1 Capacity, boundaries and predictive memory

411 For a finite state space X , Shannon entropy satisfies $H(p) \leq \log |X|$, with
 412 equality only for the uniform distribution. A stream with more possible
 413 histories than $|X|$ cannot be encoded injectively in that register. Separately,
 414 a continuous field on a connected substrate that reaches distinct components
 415 of a disconnected vacuum manifold must leave that manifold somewhere,
 416 the simplest topological-defect construction (Kibble, 1976). Relating this
 417 locus to a particular finite information-processing register is the additional
 418 hypothesis E12 carries.

419 A non-surjective update on a finite register is non-injective. Under the
 420 specified system–bath model and thermal assumptions, erasure carries a Lan-
 421 dauer heat cost (Landauer, 1961). Finite capacity does not imply that every
 422 update erases, and logical many-to-one processing must be distinguished from
 423 its physical implementation (Norton, 2005; Shenker, 2000). The formaliza-
 424 tion records the physical heat identity as an obligation of each model and
 425 supplies an explicit one-bit eraser satisfying it.

426 For a system state X_t driven by an environmental signal S_t , the predictive

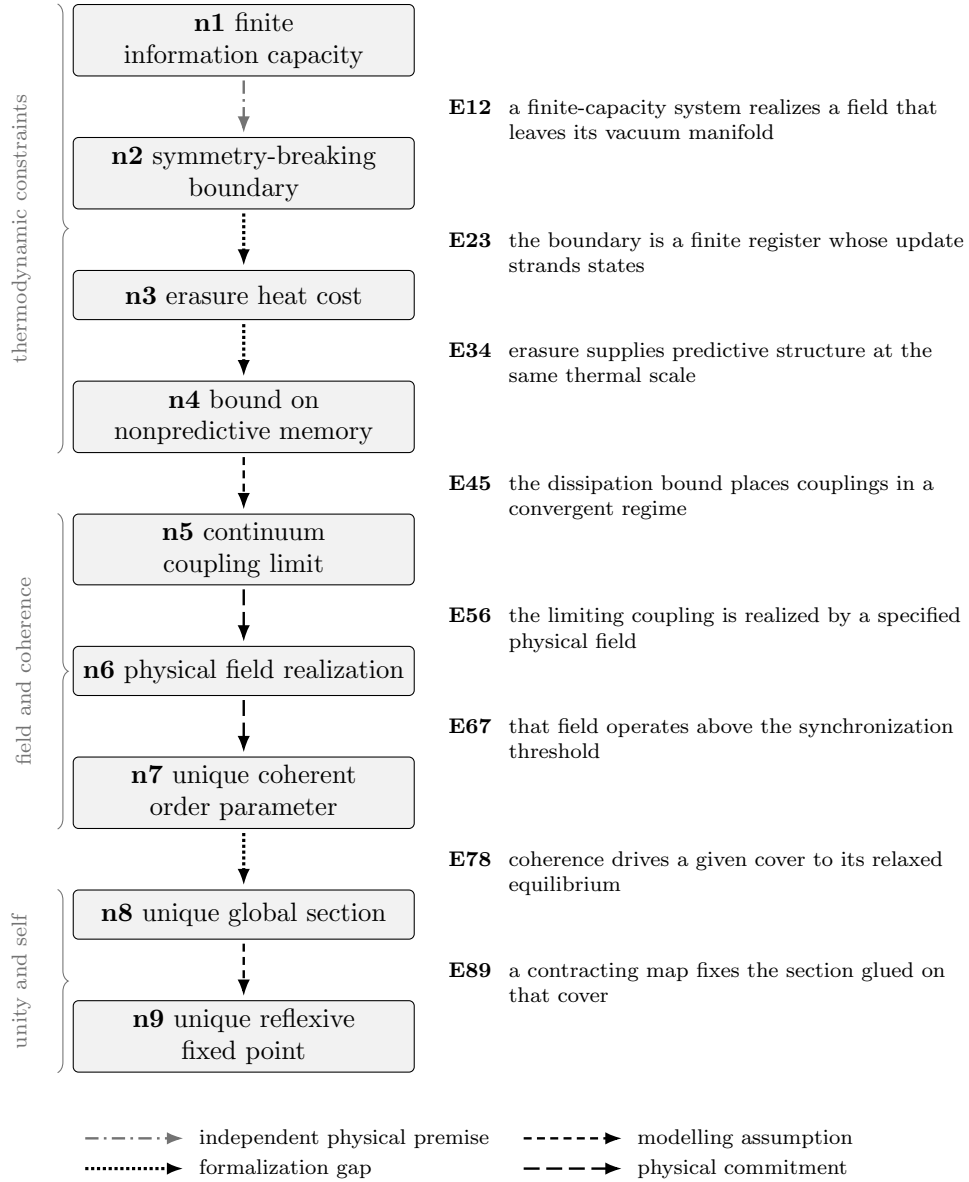


Figure 4: The conditional composition. Each node is an established or cited result; each arrow is an implication the composition consumes and no theorem here supplies, styled by the kind of assumption it carries. The three formalization gaps bracket the two physical commitments, so the chain’s empirical content sits between links of a different kind on either side. E12 is drawn as a premise because n1 and n2 are independent roots. Table 1 states each connection in full.

Edge	Status	Required connection
E12	Independent	A finite-capacity system realizes a field that leaves its vacuum manifold. Capacity alone does not imply this.
capacity $\rightarrow$ boundary	physical premise	
E23	Formalization	The boundary is associated with a finite register whose update leaves states unreachable.
boundary $\rightarrow$ erasure	gap	
E34	Formalization	Erasures supplies a predictive structure with the same thermal scale and heat budget, and nonzero wasted memory.
erasure $\rightarrow$ prediction	gap	
E45	Modelling	A predictive dissipation bound places coupling energies in a convergent coarse-graining regime.
prediction $\rightarrow$ continuum	assumption	
E56	Physical	The limiting coupling is realized by a specified field satisfying the kernel conditions. Its identification with cortex is empirical.
continuum $\rightarrow$ field	commitment	
E67	Physical	That field operates above the synchronization threshold.
field $\rightarrow$ coherence	commitment	
E78	Formalization	Coherence drives a cover, supplied as a separate argument of the theorem, to an equilibrium reached by relaxation at coupling at least the named mean-field value. That its patches carry compatible local states is a field of the cover rather than a conjunct of this edge.
coherence $\rightarrow$ unity	gap	
E89	Modelling	The state glued on that same cover is fixed by a contracting map mediated by its local self-encoding.
unity $\rightarrow$ self	assumption	

Table 1: The assumptions consumed by the conditional composition theorem, each an implication between two of the nine nodes above. Node-level results do not establish these connecting implications. The complete theorem map and witnesses are in Supplemental Material, Table S1.

427 dissipation bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (7)$$

428 Under the no-back-action Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$, data process-
429 ing makes the bracket nonnegative (Still et al., 2012). The bound constrains
430 nonpredictive memory at a given dissipation budget. It does not specify
431 a learning rule, maximize prediction, or require the internal representation
432 to reproduce environmental geometry—the point the plasticity control of
433 Section 4 makes concretely. Connecting this budget to Landauer heat ad-
434 ditionally requires nonpredictive information to be bounded by the entropy
435 erased by the register. The formal bridge derives Eq. (7) from Landauer’s
436 bound under that identification; it does not derive the identification.

437 A bound $D_{KL} \leq \Delta t \sigma$ also cannot force $D_{KL} \rightarrow 0$ unless its right-hand
438 side tends to zero, and Gibbs’ inequality proves nonnegativity of relative en-
439 tropy, not a general identification of relative entropy with physical entropy
440 production. The chain uses a different technical sense of information at each
441 step—a counting bound on a register, a thermodynamic heat cost, a predic-
442 tive mutual information—and those senses are not interchangeable (Rosas
443 et al., 2026). These distinctions prevent an informational inequality from
444 being read as a derivation of environmental learning, Bayesian inference, or
445 semantic content.

446 5.2 Compatible states and reflexive fixed points

447 A sheaf formalizes local descriptions and their restriction to smaller regions.
448 If local sections agree on every overlap, the sheaf condition gives a unique
449 global section. In the probability-sheaf construction, this is consistency of
450 local density information, and it is not implied by agreement of oscillator
451 phases: the development includes a counterexample with equal phases but
452 incompatible densities on an overlap. Section 6 rests on this distinction,
453 and the model excludes such configurations only through its compatibility
454 hypothesis.

455 A thermodynamic cover couples this gluing construction to a phase con-
456 figuration at a potential minimum. For the deterministic positive-coupling
457 model, convergence to that minimum is proved under an explicit initial basin
458 condition; other equilibria prevent an unconditional conclusion. The station-
459 ary noisy order parameter, the deterministic relaxation, and the cover must
460 therefore be connected by E78 rather than identified without a hypothesis.
461 Neither field continuity nor synchronization alone guarantees a globally com-
462 patible representation.

463 A reflexive extension equips global sections with a complete metric and a
464 predictive map formed by restricting a state to a local encoding region and
465 reading that encoding out. If the map contracts and fixes the section supplied
466 by the cover, Banach's theorem gives uniqueness of that same fixed point.
467 The contraction rate is assumed, and the equality between the glued section
468 and the fixed point is part of E89. A coherent order parameter supplies the

469 supercritical inequality required by the stated contraction bound; it does not
470 construct the map or prove that cortical dynamics implements it.

471 The theorem `chain` concludes `UnifiedSelf` from these hypotheses and
472 from two structures it also takes as arguments, a thermodynamic cover and
473 a reflexive boundary. The eight edges are not the whole of what is assumed:
474 the cover’s overlap-compatibility field—the load-bearing assumption of the
475 unity-is-not-synchrony argument—and the boundary’s resonance condition
476 are class fields, so they are consumed by the theorem without appearing
477 among its edges. A single witness satisfies all eight at once, so their conjunc-
478 tion is not empty; it is assembled from a double well, a register, a uniform-grid
479 mesh and a three-site cortex, which remain distinct toy systems rather than
480 one mechanism. The project declares no additional Lean axioms beyond
481 the foundational axioms used by Lean’s libraries; physical content resides
482 in hypotheses and model fields. This checks the conditional mathematical
483 composition, not the empirical truth of its premises. Detailed constructions,
484 scope statements and counterexamples are retained in the supplement and
485 source files.

486 **6 What the construction commits to about** 487 **consciousness**

488 The composition terminates in a mathematical object, and we take a posi-
489 tion on what that object is. The commitment has two clauses. *Unity*: the

490 unique global section glued from compatible local states is the formal corre-
491 late of the unity of a conscious episode—the fact that a subject undergoes
492 one experience rather than a collection of co-occurring local ones. *Selfhood*:
493 the fixed point of the self-prediction map on that same section is the formal
494 correlate of a minimal self, a state that is what it takes itself to be. Neither
495 clause is a theorem, and no simulation here supports either. We argue for
496 them, then say what would count against them.

497 Three features of the construction motivate the identification, and each
498 distinguishes it from a bare synchrony measure.

499 First, compatibility rather than agreement. Oscillator phases can agree
500 everywhere while the local probabilistic descriptions they carry remain in-
501 compatible on their overlaps, and Section 5.2 exhibits that configuration
502 inside the formalization, not only as an illustration. If the unity of experi-
503 ence is a matter of local contents being about one world rather than merely
504 co-timed, then compatibility, not coherence, is the property to look for—and
505 coherence enters as the physical condition under which compatibility can be
506 maintained, which is what E78 asserts.

507 Second, uniqueness follows from the construction rather than from a sep-
508 arate selection principle. Accounts of conscious unity need something that
509 picks one experience from among the candidates a system could support;
510 integrated information theory postulates exclusion for this purpose (Tononi
511 et al., 2016). Here uniqueness arrives twice as a consequence: given a cover
512 whose local sections agree, the sheaf condition gives exactly one global sec-

tion, and given a contracting self-prediction map, Banach’s theorem gives exactly one fixed point. The scope of that claim needs stating exactly, because the stronger reading is available and is not ours. Uniqueness is relative to a cover and a map, and choosing them—which regions, which overlaps, which encoding region, and so where the subject ends—is not supplied by either theorem. It is what E78 and E89 carry. What the construction removes is the need for an additional principle to select among the experiences a fixed cover and map would otherwise admit; it does not remove the need to say which cover and map a given subject has. That is a structural argument, not evidence.

Third, the reflexive clause makes the self a condition rather than an addition. The predictive map restricts a global state to a region encoding it and reads that encoding back out, so a fixed point is a state whose self-model is accurate about the state it models. That condition is available only because the state modelled is a single glued object: a system whose local descriptions do not cohere has no such state for a self-model to be about. Unity and selfhood are therefore one construction read at two levels, which is why `UnifiedSelf` requires the fixed point to be the glued section itself rather than an unrelated section.

The costs belong with the argument. The commitment does not address why any physical condition is accompanied by experience; it is a proposal about which physical state is the conscious one, not about why states of that kind feel like anything. It is therefore open to the objection that a system

536 could satisfy every condition—compatible local states, a supercritical field,
537 an accurate self-model—while nothing is experienced. We have no reply to
538 that objection and claim none. The identification is also multiply realizable
539 as stated: nothing in the sheaf or fixed-point conditions is specific to cortex
540 or to electromagnetic coupling, so the substrate claim is carried entirely by
541 E56 and E67 and is empirical. No conclusion about artificial systems follows
542 from the formal results.

543 What could count against the commitment is dissociation. It predicts that
544 the two clauses and the presence of a unified conscious episode go together,
545 so a state in which local cortical descriptions are demonstrably compatible
546 and the field supercritical, while the subject reports no unified experience,
547 would tell against it; so would unified reported experience where compati-
548 bility demonstrably fails. Anesthetic emergence, dreaming and dissociative
549 states are the places to look, because they separate coherence from report
550 in ways ordinary waking does not. Such tests need an operational measure
551 of overlap compatibility on neural data, which this article does not supply;
552 constructing one is the main obstacle to making the commitment empirically
553 live rather than merely stated. Supplemental Material specifies what such an
554 observable would have to satisfy—a restriction map, a statistic with a null
555 under decoding noise, validation against configurations whose compatibility
556 is built rather than inferred—and the three ways it would fail silently, one
557 of which is reading compatibility off synchrony and so assuming away the
558 distinction being tested.

559 The physical tests of Sections 3 and 4 are independent of all of this. They
560 concern coupling, coherence and recovery, and can fail while the identification
561 stands, or succeed while it is wrong.

562 **7 Discussion**

563 The framework connects thermodynamic information constraints, collective
564 phase organization and compatible local representations, conditionally and
565 with its assumptions named. Its biological content is a coupling-gated re-
566 covery hypothesis with defined observables, a stated protocol, and failure
567 conditions declared in advance. Its formal content is a composition whose
568 eight gaps are exhibited rather than argued past, and which can be inhabited
569 by a single model. Its interpretive content is the identification of Section 6,
570 offered as a commitment with reasons.

571 The simulations are useful because they restrict the hypothesis rather
572 than support it. Finite ramps smear the stationary onset to the point where
573 a fitted onset exponent chiefly reports the ramp; an unpinned collective
574 phase makes independence diagnostics ambiguous in exactly the supercritical
575 regime the model cares about; and squared-drift plasticity maintains coher-
576 ence while degrading structural alignment. The second and third of these
577 are results about method and about a common intuition, and they transfer
578 beyond this framework.

579 The proposal differs in emphasis from accounts based on integrated in-

580 formation (Tononi et al., 2016), global workspace dynamics (Baars, 1997;
581 Dehaene et al., 2014), or predictive processing (Friston, 2010). It specifies a
582 candidate interaction medium and a mathematical compatibility condition,
583 and, relative to a given cover and self-prediction map, it derives the unique-
584 ness those accounts postulate. It does not derive the exclusion principle of
585 integrated information, identify every broadcast with a field transition, or
586 derive a generative model from a dissipation bound. Continuous fields can
587 carry incompatible local descriptions and discrete models can admit sheaf
588 constructions, so continuity alone does not settle the combination problem;
589 compatibility is what this account offers in its place.

590 Two formal results bear on the substrate question, and both admit a
591 stronger reading than they support. At matched coupling resource, a fixed
592 wiring support that excludes a better-correlated pair can be suboptimal—a
593 statement about constrained optimization, not about biology. On an atom-
594 less substrate, a kernel supported on finitely many points contributes zero to
595 the chosen continuum coupling integral. That is a property of the measure
596 and the observable, and it cuts against this framework’s own coarse-graining
597 step as much as against anything else. It does not imply that digital hard-
598 ware cannot approximate a field or support consciousness, and no biological-
599 versus-artificial conclusion follows.

600 The empirical priorities follow directly from Section 3.2: estimate effec-
601 tive coupling and diffusion in consistent units, test the proposed relation
602 between coupling and extracellular geometry, and measure recovery against

an independently defined behavioral endpoint under matched model comparison. A favorable concentration–coherence relation would support a statistical description, not identify its physical cause; an unfavorable result in a regime satisfying the stated assumptions would constrain that description. Successful synchronization simulations cannot substitute for measurements distinguishing ephaptic interactions from other coupling mechanisms.

The article therefore asks to be judged in three separable parts: a prediction that can be tested, a composition that can be checked, and an identification that can be argued about. Keeping them separate is what makes the whole assessable.

Data and code availability

The Lean development, simulation scripts, tests and saved numerical summaries are maintained in the manuscript source repository. Publication simulation values are regenerated from saved summaries without rerunning production sweeps. The exploratory EEG analysis uses OpenNeuro dataset ds005620 (Bajwa et al., 2025); preprocessing and selection are described in Supplemental Material. The source repository contains the reproducible analysis and does not redistribute the raw recordings.

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622 This research received no external funding. The author declares no compet-
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624 **Use of AI tools**

625 AI assistance to this work was substantial and is not separable by task.
626 Several coding assistants and chat interfaces were used in an interleaved way
627 throughout: Google’s Antigravity CLI (Gemini 3.1, Claude 4.6 Opus, Claude
628 4.6 Sonnet), OpenAI’s Codex CLI (Astra, Terra, Luna, Sol), Claude Code
629 (Claude Opus 5), Gemini Pro through its web interface, and Fable (Claude
630 Opus 4) through OpenRouter, accessed with the Hermes CLI. Between them
631 they contributed to the Lean development, the simulation and analysis code,
632 and the drafting and revision of both documents. Because the tools were
633 used in combination and in alternation, no section, proof, figure or result is
634 attributable to any one of them, and none is claimed to be. The author is
635 responsible for all content, including everything a tool produced. Machine
636 checking establishes the stated Lean results under their hypotheses; it does
637 not validate biological interpretations or replace scientific review of generated
638 code and prose.

References

- Pinotsis, D. A., & Miller, E. K. (2022). Beyond dimension reduction: Stable electric fields emerge from and allow representational drift. *NeuroImage*, 253, 119058.
- Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated information theory: from consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17(7), 450-461.
- Baars, B. J. (1997). In the Theater of Consciousness: The Workspace of the Mind. *Oxford University Press*.
- Dehaene, S., Charles, L., King, J. R., & Marti, S. (2014). Toward a computational theory of conscious processing. *Current Opinion in Neurobiology*, 25, 76-84.
- Friston, K. (2010). The free-energy principle: a unified brain theory?. *Nature Reviews Neuroscience*, 11(2), 127-138.
- Pinotsis, D. A., & Miller, E. K. (2023). In vivo ephaptic coupling allows memory network formation. *Cerebral Cortex*, 33(17), 9877-9890.
- Pinotsis, D. A., Fridman, G., & Miller, E. K. (2023). Cytoelectric coupling: Electric fields sculpt neural activity and "tune" the brain's infrastructure. *Progress in Neurobiology*, 226, 102465.

- 658 McFadden, J. (2002). The conscious electromagnetic information (cemi) field
659 theory: the hard problem made easy?. *Journal of Consciousness Studies*,
660 9(8), 45-60.
- 661 Pockett, S. (2012). The electromagnetic field theory of consciousness: a
662 testable hypothesis about the characteristics of conscious as opposed to
663 non-conscious fields. *Journal of Consciousness Studies*, 19(11-12), 191-223.
- 664 McFadden, J. (2020). Integrating information in the brain's EM field: the
665 cemi field theory of consciousness. *Neuroscience of Consciousness*, 2020(1),
666 niaa016.
- 667 Fröhlich, F., & McCormick, D. A. (2010). Endogenous electric fields may
668 guide neocortical network activity. *Neuron*, 67(1), 129-143.
- 669 Anastassiou, C. A., & Koch, C. (2015). Ephaptic coupling to endogenous
670 electric field activity: why bother?. *Current Opinion in Neurobiology*, 31,
671 95-103.
- 672 Pockett, S. (2002). Difficulties with the electromagnetic field theory of con-
673 sciousness. *Journal of Consciousness Studies*, 9(4), 51-56.
- 674 Vöröslakos, M., et al. (2018). Direct effects of transcranial electric stimulation
675 on brain circuits in rats and humans. *Nature Communications*, 9, 483.
- 676 Verkhratsky, A., & Nedergaard, M. (2018). Physiology of Astroglia. *Physio-*
677 *logical Reviews*, 98(1), 239-389.

678 Bastos, A. M., Donoghue, J. A., Brincat, S. L., Mahnke, M., Yanar, J., Cor-
679 rea, J., ... & Miller, E. K. (2021). Neural effects of propofol-induced uncon-
680 sciousness and its reversal using thalamic stimulation. *eLife*, 10, e60824.

681 Bajwa, I. J., Nilsen, A. S., Skukies, R., Aamodt, A., Ernst, G., Storm, J.
682 F., & Juel, B. E. (2025). A repeated awakening study exploring the capac-
683 ity of complexity measures to capture dreaming during propofol sedation.
684 *Scientific Reports*, 15, 32012.

685 Xie, L., Kang, H., Xu, Q., Chen, M. J., Liao, Y., Thiyagarajan, M., ... &
686 Nedergaard, M. (2013). Sleep drives metabolite clearance from the adult
687 brain. *Science*, 342(6156), 373-377.

688 Norton, J. D. (2005). Eaters of the lotus: Landauer's principle and the return
689 of Maxwell's demon. *Studies in History and Philosophy of Science Part B:*
690 *Studies in History and Philosophy of Modern Physics*, 36(2), 375-411.

691 Landauer, R. (1961). Irreversibility and heat generation in the computing
692 process. *IBM Journal of Research and Development*, 5(3), 183-191.

693 Kibble, T. W. B. (1976). Topology of cosmic domains and strings. *Journal*
694 *of Physics A: Mathematical and General*, 9(8), 1387-1398.

695 Sakaguchi, H. (1988). Cooperative phenomena in coupled oscillator systems
696 under external fields. *Progress of Theoretical Physics*, 79(1), 39-46.

697 Still, S., Sivak, D. A., Bell, A. J., & Crooks, G. E. (2012). Thermodynamics
698 of prediction. *Physical Review Letters*, 109(12), 120604.

- 699 Kawai, R., Parrondo, J. M. R., & Van den Broeck, C. (2007). Dissipation:
700 The phase-space perspective. *Physical Review Letters*, 98(8), 080602.
- 701 Corberi, F., dello Russo, S., Messuti, G., Scarpetta, S., & Smaldone, L.
702 (2026). Thermodynamic learning. *arXiv preprint*. arXiv:2609.04732.
- 703 Banach, S. (1922). Sur les opérations dans les ensembles abstraits et leur ap-
704 plication aux équations intégrales. *Fundamenta Mathematicae*, 3(1), 133-
705 181.
- 706 Shenker, O. R. (2000). Logic and entropy. *PhilSci-Archive preprint*. philsci-
707 archive.pitt.edu/115.
- 708 Rosas, F. E., Mediano, P. A. M., Luppi, A. I., Boyd, A. B., Berta, M., Jensen,
709 H. J., Gastpar, M., & Carroll, S. M. (2026). Four faces of information in
710 natural systems. *Nature Reviews Physics*. doi:10.1038/s42254-026-00971-4.
- 711 Sznitman, A.-S. (1991). Topics in propagation of chaos. In *École d'Été de*
712 *Probabilités de Saint-Flour XIX—1989*, Lecture Notes in Mathematics,
713 vol. 1464, pp. 165–251. Springer. doi:10.1007/BFb0085169.
- 714 Townsend, R. G., Solomon, S. S., Chen, S. C., Pietersen, A. N. J., Martin,
715 P. R., Solomon, S. G., & Gong, P. (2015). Emergence of complex wave
716 patterns in primate cerebral cortex. *The Journal of Neuroscience*, 35(11),
717 4657-4662. doi:10.1523/JNEUROSCI.4509-14.2015.
- 718 Xu, Y., Long, X., Feng, J., & Gong, P. (2023). Interacting spiral wave pat-
719 terns underlie complex brain dynamics and are related to cognitive pro-

720 cessing. *Nature Human Behaviour*, 7(7), 1196-1215. doi:10.1038/s41562-
721 023-01626-5.